

XML CODE

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etBookName"
        android:hint="Enter Book Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etAuthorName"
        android:hint="Enter Author Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/btnAdd"
        android:text="Add Book"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="15dp"/>

    <Button
        android:id="@+id/btnShow"
        android:text="Show Books"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textSize="16sp"/>

</LinearLayout>
```

Data Class

```
package com.example.ca
```

```
data class Book(  
    var bookName: String? = null,  
    var authorName: String? = null  
)
```

ActivityMain.kt

```
package com.example.ca
```

```
import android.os.Bundle  
import android.widget.*  
import androidx.appcompat.app.AppCompatActivity  
import com.google.firebase.database.*
```

```
class MainActivity : AppCompatActivity() {  
  
    lateinit var etBook: EditText  
    lateinit var etAuthor: EditText  
    lateinit var btnAdd: Button  
    lateinit var btnShow: Button  
    lateinit var tvResult: TextView  
  
    lateinit var database: DatabaseReference  
  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContentView(R.layout.activity_main)  
  
        etBook = findViewById(R.id.etBookName)  
        etAuthor = findViewById(R.id.etAuthorName)  
        btnAdd = findViewById(R.id.btnAdd)  
        btnShow = findViewById(R.id.btnShow)  
        tvResult = findViewById(R.id.tvResult)  
  
        database = FirebaseDatabase.getInstance().getReference("Books")  
  
        btnAdd.setOnClickListener {
```

```

val bookName = etBook.text.toString()
val authorName = etAuthor.text.toString()

if (bookName.isEmpty() || authorName.isEmpty()) {
    Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show()
} else {

    val bookId = database.push().key!!
    val book = Book(bookName, authorName)

    database.child(bookId).setValue(book)
        .addOnSuccessListener {
            Toast.makeText(this, "Book Added Successfully", Toast.LENGTH_SHORT).show()
            etBook.text.clear()
            etAuthor.text.clear()
        }
        .addOnFailureListener {
            Toast.makeText(this, "Failed to Add", Toast.LENGTH_SHORT).show()
        }
    }
}

btnShow.setOnClickListener {

    database.addValueEventListener(object : ValueEventListener {
        override fun onDataChange(snapshot: DataSnapshot) {

            val builder = StringBuilder()

            for (data in snapshot.children) {
                val book = data.getValue(Book::class.java)
                builder.append("Book: ${book?.bookName}\n")
                builder.append("Author: ${book?.authorName}\n\n")
            }

            tvResult.text = builder.toString()
        }

        override fun onCancelled(error: DatabaseError) {
            Toast.makeText(this@MainActivity, "Error", Toast.LENGTH_SHORT).show()
        }
    })
}

```

```
}  
}
```

